Representative examples of GAT layers.

For one directed edge $j \to i$, we can think GATv2 doing 3 things:

(1) Build the message for node $j$

So node $j$ has hidden representation $h_j^{(l)}$ at layer $l$.

the information it sends is a learned transformation of that representation

$$\left[ m_{j \to i}^{(l)} = W_m^{(l)} h_j^{(l)} \right] \text{ so } m_{j \to i} \text{ means "the info node } j \text{ is sending to } i"$$

(2) compute how important that sent information is to node $i$? $\Rightarrow$ through score $s_{ij}$

the attention mechanism will need the hidden representation $h_j^{(l)}$ and $h_i^{(l)}$

as well $e_{ij} = [\Delta x, \Delta y, d]$ and with these it will compute a score $s_{ji}^{(l)}$

$$s_{ji}^{(l)} = a^T \text{leakyRelu}\left( W_i h_i^{(l)} + W_j h_j^{(l)} + W_e e_{ji} \right) \text{ then we normalize it use Softmax}$$

to get the attention weights $\alpha_{ji}^{(l)} = \text{softmax}\left( s_{ji}^{(l)} \right)$ so

$\alpha_{ji}^{(l)} \to$ "how much node $i$ should care about message from node $j$" when updating

(3) update node (1) $\Rightarrow$ use weighted average just weighted sum.

$$\left[ h_i^{(l+1)} = \sum_{j \in N(i)} \alpha_{ji}^{(l)} m_{j \to i}^{(l)} \right] \text{ or since } m_{j \to i} = W_m^{(l)} h_j^{(l)} \text{ we can write}$$

$$\hookrightarrow h_i^{(l+1)} = \sum_{j \in N(i)} \alpha_{ji}^{(l)} W_m^{(l)} h_j^{(l)} \quad \text{where } \sum \alpha_{ji} = 1 \text{ because softmax}$$

from our Problem a $L = 2$ so $h_{i \ldots N}^{(0)} \to h_{i \ldots N}^{(1)} \to \left( h_{i \ldots N}^{(2)} \right) \Leftrightarrow H^{(2)} \in \mathbb{R}^{N \times 64}$

from $H^{(2)}$ we only take $h_0^{(2)} \in \mathbb{R}^{64}$ but for one $b$ sample. For Batch $B$ is $h_0^{(2)} \in \mathbb{R}^{B \times 64}$

[Deterministic] $\leftarrow \hat{y} = [B, 12, 2] \leftarrow$ & $h_{dec} \in \mathbb{R}^{B \times 24} \leftarrow$ pass decoder